

Enhanced Photocatalytic Activity of TiO₂ Anatase

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Abstract

We investigate the band gap engineering of TiO₂ anatase nanoparticles through nitrogen doping. DFT calculations show that nitrogen doping significantly narrows the band gap, enhancing visible-light photocatalytic activity.

Introduction

Titanium dioxide remains the most widely studied photocatalyst due to its stability, low cost, and non-toxicity. However, its wide band gap limits its photocatalytic activity to ultraviolet light.

Methods

DFT calculations were performed using VASP with PAW pseudopotentials. A 2x2x1 supercell of anatase TiO₂ was modeled, with nitrogen atoms substituting oxygen atoms at various concentrations.

Results

The calculated density of states shows a clear band gap reduction upon nitrogen incorporation. The band gap narrows from approximately 3.2 eV for pure TiO₂ to 2.4 eV for the nitrogen-doped sample.

Conclusion

Nitrogen-doped TiO₂ anatase shows significantly enhanced visible-light photocatalytic activity, validated by DFT calculations. This finding suggests a promising strategy for improving the efficiency of TiO₂ photocatalysts.